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Comprehensive Exams

Doctor of Education

Higher Education - Technology and Learning

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QUESTION 3

Part A: Quantitative Research Design (Be sure to answer all parts of the question)

A research team asks you to help design and implement a Quantitative study to investigate whether there are statistically significant differences in test scores obtained from students who have in their classes been exposed to three different types of educational environments over the course of an academic year. One group from school A receives traditional educational instruction, another group from school B receives instruction enhanced by additional work with computers and the internet, while a third group from school C receives instruction enhanced by additional training in multi-cultural sensitivity. Approximately 60 participants will be obtained from each school. Participants will include only those whose parents or guardians grant permission for use of student records.

1. Identify and define the following:
 - a. the type of research design to be employed in the study
 - b. the independent and dependent variables
 - c. the type of sample to be collected.
2. Assuming that the dependent variable is measured on an interval scale, what statistical procedure (e.g. t-test, chi-square, etc.) would be most appropriate to evaluate the research question?
3. If the correct statistical test used to evaluate these data yields a p value (or the Sig. value in SPSS) of .040, what does that result indicate about the groups?

4. If the correct statistical test used to evaluate these data yields a p value (or the Sig. value in SPSS) of .895, what does that result indicate about the groups?

Part B: Qualitative Research Design (Be sure to answer all parts of the question)

A research team asks you to help design and implement a Qualitative study to investigate similarities and differences in perceptions of performance of students who have in their classes been exposed to three different types of educational environments over the course of an academic year. One group from school A receives traditional educational instruction, another group from school B receives instruction enhanced by additional work with computers and the internet, while a third group from school C receives instruction enhanced by additional training in multi-cultural sensitivity. Participants will include only those whose parents or guardians grant permission for use of student records.

1. Describe in detail how you would:

- a. Design your study
- b. Collect data
- c. Analyze data

2. Describe how each of the following key terms apply in the design of your study:

- a. Bracketing or Reflexivity
- b. Open Coding
- c. Axial Coding
- d. Triangulation

e. A gatekeeper Emic reporting

g. Etic reporting

Remember to include a bibliography of sources cited in your answer in APA format. The maximum length for Question 3 is 10 pages, excluding references.

Question 3

Part A: Quantitative Research Design

1. Identify and define the following:

The purpose of this study is to investigate whether there are statistically significant differences in test scores obtained from students who have, in their classes, been exposed to three different types of educational environments (traditional educational instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multi-cultural sensitivity) over the course of an academic year.

Research Design

A quantitative experimental design will be employed, and participants will be randomly assigned to treatment and control groups. Traditional educational instruction is the control group. Instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multicultural sensitivity, are the experimental groups (Gay, Mills & Airasian, 2006, p. 254).

Variables

The independent variable denotes different treatment conditions, with different groups receiving different treatments. In this study, there are three independent group variables, and they are defined by the “type of educational environment”: a) 1 = traditional; b) 2 = computer and the Internet; and c) 3 = multicultural sensitive.

The dependent variable is a variable that denotes a measurement on a coherent, scaled attribute, as defined by a score on that attribute. In this study, the dependent variable is defined as the “test score”.

Sample

Approximately 60 participants (i.e., students) will be randomly selected from each school. Participants will include only those whose parents or guardians grant permission for the use of student records. In this study, approximately 180 participants will be needed to carry out the research.

2. Assuming that the dependent variable is measured on an interval scale, what statistical procedure would be most appropriate to evaluate the research question?

The statistical procedure for comparing three independent groups (i.e., type of educational environment) on the dependent variable (i.e., test score) is called one-way ANOVA in SPSS. This is simply comparing the means of groups in order to make inferences about the population. One-way ANOVA is also called single-factor analysis of variance because there is only one independent variable or factor. The independent variable has nominal levels (Gay et al., 2006, p. 359; Morgan, Leech, Gloeckner, & Barrett, 2007, p. 157). In this study, we will use the one-way ANOVA procedure because we have one independent variable with three independent groups.

The level of measurement of the independent variable is nominal because there are three unordered categories (i.e., types of educational environments), and numerical values are used only to uniquely identify the attribute.

The dependent variable is measured on an interval scale, in which distances between levels are equal but there is no true zero.

3. If the correct statistical test used to evaluate these data yields a p-value (or the Sig. value in SPSS) of .040, what does that result indicate about the groups?

Either 5 out of 100 or .05 is the typical alpha level that researchers use to assess whether the null hypothesis should be rejected or not (Gay et al., 2006, p. 343; Morgan et al., 2007, p. 92).

For $p = .04$, the result of this study is unlikely to be due to chance alone, because the p value is less than .05 (i.e., $p < .05$). The result suggests that there is a significant difference among the three types of educational environments (or treatment conditions) in test scores. However, we do not know which specific groups of means are significantly different unless we do follow-up tests. Multiple comparisons can be used to determine which means are significantly different from one another (Morgan et al., 2007, p. 160).

4. If the correct statistical test used to evaluate these data yields a p -value (or the Sig. value in SPSS) of .895, what does that result indicate about the groups?

For $p = .895$, the result of this study is likely due to chance alone, because p value is larger than .05 (i.e. $p > .05$). The result suggests that there is not a significant difference in average student test scores among the three types of educational environments (or treatment conditions) and that there is no evidence to suggest the difference and no further test is needed.

Part B: Qualitative Research Design

Research Design

Purpose

The purpose of this study is to investigate similarities and differences in students' perceptions of performance among those who, in their classes, have been exposed to three different types of educational environments over the course of an academic year.

Research Question

Are there similarities and differences between the types of environments (traditional educational instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multi-cultural sensitivity) in regard to the perceptions of performance of students?

Method

A qualitative experimental design will be employed. Phenomenological methodology will be used to investigate similarities and differences between the types of environments with regard to perceptions of students' performance. This method reveals the meaning of our everyday existence. According to van Manen (1990), it is "the fulfillment of our human nature: to become more fully who we are" (p. 12). This perspective suggests that individuals reflect many of their sentiments regarding research inquiries and data analysis that observe individuals' lives within their unique contexts (Creswell, 2007, p. 69).

In this study, the investigator/researcher (etic reporting) collects descriptions of behavior through observations, interviews, documents, and artifacts (Creswell, 2007, p. 131).

Site and Participant Selection

In this study, we will stratify a sample of 10 students by class from each of the three types of educational environments (traditional instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multicultural sensitivity). The larger sample size may be used to ensure that smaller (or harder-to-reach) subgroups are still likely to be represented. However, a larger sample size may waste time, resources, and money. Creswell (2007) stated that in phenomenology, he saw the number of

participants range from 1 to 325 (p. 126). In this study, emic reporting is the participant themselves.

A gatekeeper may be used to contact potential participants for the study. Creswell (2007) suggested “bracketing” prior to the interview process.

Consent Form

To protect human subjects, all participants must have their parents or guardians review and sign a consent form prior to their participation in the study.

Interview Questions

In this study, we will ask all participants three open-ended questions across the three types of educational environments. The following questions are addressed:

1. How would you describe your school’s environment?
2. What do you already know about multicultural diversity?
3. What kinds of learning experiences do you have in your classroom, such as art, music, math, science, social studies, computers, and the Internet?

Analyze data

Phenomenological data analysis tends to follow specific structured methods. Creswell (2007) suggests three general approaches to analysis—Madison, Huberman and Miles, and Wolcott. In this study, we will use data analysis strategies by Wolcott, which suggest sketching ideas, reducing codes to themes, relating categories to an analytic framework in the literature, and displaying the data (Creswell, 2007, p. 149).

In addition, researchers can use several techniques discussed by Patten (2005) to establish the data's dependability and trustworthiness. Specific techniques include a) data triangulation; b)

methods triangulation; c) researcher triangulation; d) team of researchers with diverse backgrounds; e) interobserver agreement; f) peer review; g) auditor; and h) member checking.

In this study, we will use inductive analysis, which means that categories and patterns emerge from the data rather than being imposed on it prior to data collection (McMillan, 2001, p. 462). We will also use open coding and axial coding. In addition, we will use triangulation to establish the dependability and trustworthiness of the data.

Before we begin analyzing the data, we will need to transcribe the interviews. Initially, we will use hand coding of the transcripts using paper-and-pencil methods, and a codebook will be constructed, revised, and used throughout the data analysis to record emerging concepts and subcategories.

Qualitative data includes structured text, unstructured text, audio recordings, video recordings, etc. Creswell (2007) suggests that Computer-Assisted Qualitative Data Analysis Software (CAQDAS) is very helpful in computerizing the analysis of text and image data (p. 168). Creswell goes on to list the following advantages of using computers as a tool to accomplish qualitative research:

1. Provides an organized storage system.
2. Can help researchers locate material quickly and easily.
3. Encourages researchers to look closely at the data line by line.
4. The concept mapping feature enables the researcher to visualize relationships by drawing a visual model.
5. Allows a researcher to easily retrieve memos associated with codes, themes, or documents (Creswell, 2007, p. 165).

2. Describe how each of the following key terms apply in the design of your study:

Bracketing or Reflexivity

Bracketing, reflectivity, or epoche are concepts in which researchers set aside their experiences as much as possible to take a fresh perspective on the phenomenon under examination (Creswell, 2007, p. 59). These ideas suggest that researchers should set aside what they know about the student so that it does not influence their study results.

Open Coding

Open coding is the first step in the data analysis process. It involves taking the students' interview transcripts and segmenting them into information categories. This suggests that researchers should develop a small set of categories and gradually reduce their number until the data coalesce into themes for the study (Creswell, 2007, p. 64).

Axial Coding

Axial coding is the second step, following open coding. In this step, researchers take the categories from open coding, identify one as a central phenomenon among the types of environments, and then return to the database to identify the cause of the phenomenon (Creswell, 2007, p. 237). This suggests that categories are systematically developed and linked with subcategories.

Triangulation

This is the process of using multiple data collection methods and sources to obtain a more complete picture of the subject under study and to cross-check information. The strength of qualitative research lies in collecting information in multiple ways, rather than relying on a single method. Often, two or more methods can be used so that the weakness of one is compensated for by the strength of the other (Creswell, 2007, p. 208). For example, interviews with students may help us understand what we observe happening in a learning environment.

Gate keeper

A gatekeeper is an informant or participant who allows the researcher to observe and spend time talking with participants in the study (Creswell, 2007, p. 68). A gatekeeper is essentially a person who helps the researcher to get in contact with the students.

Emic reporting

Emic (participant) reporting is when an insider participant observes the study and recounts what happened during the interview (Creswell, 2007, p. 72).

Etic reporting

Etic (researcher) reporting is very similar to Emic reporting. Etic reporting is by an outsider participant who explains the observer's role to the participants (i.e., students) (Creswell, 2007, p. 72).

References

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